Sharp Level-Set Dimensions for the Takagi–van der Waerden Functions

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Abstract

The level sets of the Takagi–van der Waerden functions reflect a competition between flat returns and crossings that may become flat only at later scales. We prove a sharp upper bound of 1/2 for the upper box dimension of every level set, uniformly over all integer bases. The proof separates the two sources of new flat pieces by their height residues and closes a coupled counting recursion. It also gives an explicit covering bound and finite 1/2-dimensional Hausdorff measure. We construct dense families of levels attaining the bound, including uncountably many irrational levels in base three. For odd bases, we describe selected entire level sets and give a finite classification on a dense arithmetic family. All general results have analytic proofs; an accompanying exact-arithmetic script supplies finite cross-checks.

Keywords. Takagi function; van der Waerden function; level set; Hausdorff dimension; box dimension; self-affinity.

1 Introduction

For an integer $r \geq 2$, let

$$T_r(x) = \sum_{n=0}^{\infty} \frac{\psi(r^n x)}{r^n}, \qquad \psi(x) = \operatorname{dist}(x, \mathbb{Z}), \qquad x \in \mathbb{R}. \tag{1}$$

Write $M_r = \max_{[0,1]} T_r$ and $L_r(y) = \{x \in [0,1] : T_r(x) = y\}$. Throughout, $\mathbb{N} = \{1, 2, \ldots\}$ and $a \bmod b = a - b\lfloor a/b \rfloor$ for real a and $b > 0$. The case $r = 2$ is the classical Takagi function; the family includes the function studied by van der Waerden. Historical and analytic background is surveyed in [1].

The geometry of the level sets is considerably richer than the elementary construction of these functions might suggest. Baba [2] determined their maxima and maximizing sets. In even bases the maximal level set has Hausdorff dimension 1/2, whereas in odd bases the unique global maximizer is $x = 1/2$; see also [3]. Thus the odd-base maximal level set does not reveal the largest dimension of a level set. For the classical function, Maddock [4] obtained an upper bound of 0.668, and de Amo, Bhouri, Díaz Carrillo and Fernández-Sánchez [5] proved the sharp bound 1/2. Allaart [6] developed a related method that counts horizontal pieces and nonhorizontal crossings separately, including signed binary Takagi functions. In [3, Introduction], the extension of the bound to even bases was noted, while the corresponding odd-base question was explicitly left unclear.

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Another approach decomposes fibers into local level sets. Jiang, Ying and Zhang [7] extended this framework to every integer base and computed the expected number of local level sets in a randomly chosen fiber. Here we seek a uniform estimate that also controls exceptional ordinates, where the largest dimensions occur.

This paper gives a self-contained counting proof for every integer base. The difference between even and odd bases is retained in the recursion: an odd base permits contributions from two adjacent height strips. The decisive arithmetic fact is that these additional crossings cannot create flat pieces at the same height residues as flat parents. Consequently the two sources are compared by a maximum, rather than added. This prevents delayed returns from increasing the exponential covering rate.

1.1 Main results

We use the convention $\dim_{\mathrm{H}} \varnothing = \overline{\dim}_{\mathrm{B}} \varnothing = 0$. Set

$$\gamma_r = \begin{cases} 1, & r \text{ even}, \\ 2, & r \text{ odd}, \end{cases} \qquad \delta_m = r^{1-2m}, \quad I_{m,j} = \left[\frac{j\delta_m}{2}, \frac{(j+1)\delta_m}{2} \right] \quad (0 \le j < 2r^{2m-1}), \tag{2}$$

and let $N_{r,2m}(y)$ be the number of these closed intervals whose T_r-image contains y. A common endpoint is counted in each incident interval.

Theorem 1.1 (Uniform covers and dimensions)**.** *For every integer $r \ge 2$, every $y \in \mathbb{R}$ and every $m \ge 1$,*

$$N_{r,2m}(y) \le 2r^{m-1} + \frac{4(r^m - \gamma_r^m)}{r - \gamma_r} \le 14 r^{m-1}. \tag{3}$$

In particular,

$$\dim_{\mathrm{H}} L_r(y) \le \overline{\dim}_{\mathrm{B}} L_r(y) \le \frac{1}{2}, \qquad \mathcal{H}^{1/2}(L_r(y)) \le \frac{2 + 4r/(r-\gamma_r)}{\sqrt{2r}}. \tag{4}$$

Here Hausdorff measure is defined using powers of diameters, without a normalizing factor.

The covering theorem is stronger than a dimension estimate alone: its constant is uniform in the ordinate, and it controls the critical Hausdorff measure. It does not assert that this measure is positive whenever the dimension is $1/2$.

Theorem 1.2 (Sharpness and irrational ordinates)**.** *For every integer $r \ge 2$, the set $\{y \in [0, M_r] : \dim_{\mathrm{H}} L_r(y) = 1/2\}$ is dense in $[0, M_r]$, and*

$$\dim_{\mathrm{H}} L_r\left(\frac{kr}{r^2 - 1} \right) = \frac{1}{2} \quad \left(1 \le k \le \left\lfloor \frac{r}{2} \right\rfloor \right). \tag{5}$$

More generally, every flat partial-sum interval gives the family of maximal-dimensional ordinates in equation (20) below. For $r = 3$, if $E \subset \mathbb{N}$ has asymptotic density zero, then

$$y_E = \frac{3}{8} - \frac{5}{3} \sum_{j \in E} 81^{-j} \quad \textit{satisfies} \quad \dim_{\mathrm{H}} L_3(y_E) = \frac{1}{2}. \tag{6}$$

In particular, every infinite $E \subset \{j^2 : j \ge 1\}$ gives an irrational ordinate, and these choices give uncountably many distinct ordinates.

For odd bases, a further result describes entire level sets, including the points outside a chosen Cantor subset. Fix an odd $r = 2q + 1 \geq 3$ and $1 \leq k \leq q$. Put $D = r^2 - 1$ and

$$\mathcal{A}_{r,k} = \left\{ y \in [0, M_r] : \text{ for some } m \geq 0, \quad r^m y \bmod 1 \in \left\{ \frac{k}{D}, \frac{kr}{D} \right\} \right\}. \tag{7}$$

Theorem 1.3 (Odd-base arithmetic structure)**.** *The set $\mathcal{A}_{r,k}$ is dense. For $y \in \mathcal{A}_{r,k}$, the level set $L_r(y)$ is either finite or satisfies $0 < \mathcal{H}^{1/2}(L_r(y)) < \infty$. These two cases can be distinguished by a finite exact procedure given a witness m in equation (7). At $y = kr/D$, the entire level set consists of two reflected copies of a Cantor set together with countably many isolated points. Moreover,*

$$N_{r,2m}\left(\frac{kr}{D}\right) = \frac{2(3r-1)r^{m-1} - 4}{r-1} \qquad (m \geq 1). \tag{8}$$

An explicit description of $L_3(3/8)$ is given in corollary 6.4. We also prove that the largest level-set dimension of $T_r(x) + tx$ equals $1/2$ for every integer t (corollary 8.1). We do not claim a necessary and sufficient classification of all maximal-dimensional ordinates, or a result for every real slope.

1.2 The counting idea and its scope

On suitable half-r-adic intervals, P_{2m} is affine with even integral slope, and its endpoint heights lie in $\delta_m \mathbb{Z}$. After adding two terms, the refinement is governed by a fixed integer staircase W_r. Flat parents create new flat pieces at residues $1, \ldots, \lfloor r/2 \rfloor$ modulo r; nonflat parents create them only at the complementary residues. If a_m and b_m denote the maximal flat and crossing counts, respectively, this gives

$$a_{m+1} \leq \max\{r a_m, b_m\}, \qquad b_{m+1} \leq r a_m + \gamma_r b_m.$$

Since $r > \gamma_r$, a common invariant bound grows by r per two scales. The remaining tail is narrower than one height strip, so the two counts also cover the actual level set.

The method follows the flat/crossing philosophy of [6, Section 3.1]; the integer profile, residue separation and odd-base two-strip estimate are proved here for general r. The historical reference to an odd-base question identifies a precise earlier formulation, rather than asserting an exhaustive priority survey of all later work. The main proofs below use no computer calculation. The supplementary script checks finite instances and illustrative examples; its role is specified in section 9.

2 Dimension conventions and an interval construction

For a bounded $F \subset \mathbb{R}$, let $\mathcal{N}_\rho(F)$ be the least number of intervals of length at most ρ covering F. We use

$$\overline{\dim}_{\mathrm{B}} F = \limsup_{\rho \downarrow 0} \frac{\log \mathcal{N}_\rho(F)}{-\log \rho}.$$

For $s > 0$, let $\mathcal{H}^s_\rho(F)$ be the infimum of $\sum_i (\operatorname{diam} U_i)^s$ over countable covers of F by sets of diameter at most ρ, and set $\mathcal{H}^s(F) = \lim_{\rho \downarrow 0} \mathcal{H}^s_\rho(F)$. Hausdorff dimension is $\inf\{s > 0 : \mathcal{H}^s(F) = 0\}$. An arbitrary subset of $\mathbb{R}$ may be replaced in a cover by its closed interval hull, with the same diameter. The definitions imply $\dim_{\mathrm{H}} F \leq \overline{\dim}_{\mathrm{B}} F$, invariance under nondegenerate similarities, and that the Hausdorff dimension of a countable union is the supremum of the dimensions. For the last assertion, apply countable subadditivity of $\mathcal{H}^s$ above that supremum. In particular a countable set has dimension zero.

We record the elementary interval construction used for the lower bounds.

Lemma 2.1 (Uniform interval construction)**.** *Let $0 < \theta < 1$. Starting from a closed interval of length $\ell_0 > 0$, suppose every interval at step $n-1$ is replaced by $b_n \geq 2$ closed subintervals of length θ times its length, with disjoint interiors. The number b_n may depend on n, but is the same for every parent at that step. Let K be the nested intersection, $B_n = \prod_{j=1}^{n} b_j$ and $\ell_n = \ell_0 \theta^n$. Then*

$$\dim_{\mathrm{H}} K = \liminf_{n\to\infty} \frac{\log B_n}{-\log \ell_n}. \tag{9}$$

If $b_n = b$ is constant and $s = \log b/(-\log\theta)$, then $0 < \mathcal{H}^s(K) < \infty$.

Proof. The step-n cover gives the upper bound in equation (9) by taking a subsequence realizing the lower limit. Assign equal probabilities to the children of each parent, and map an infinite sequence of choices to the unique point of its nested intervals. This defines a probability measure μ supported on K. Any point belongs to at most two step-n intervals, so its mass is at most $2/B_n \to 0$. Thus endpoints have zero mass, and every step-n interval has mass $1/B_n$. An interval J of length $\rho \leq \ell_n$ meets at most three step-n intervals, because those intervals have equal length and disjoint interiors. Consequently, if $\ell_{n+1} < \rho \leq \ell_n$, then $\mu(J) \leq 3/B_n$. For s strictly below the right side of equation (9), there is a constant C_s such that $B_n^{-1} \leq C_s \ell_n^s$ for all n. Hence $\mu(J) \leq 3C_s\theta^{-s}\rho^s$ at small scales. Every interval cover then has s-cost at least $(3C_s\theta^{-s})^{-1}$, by summing the measure inequality; so $\dim_{\mathrm{H}} K \geq s$. Let s increase to the lower limit. For constant $b_n = b$, the same argument works at the critical exponent, while $B_n \ell_n^s = \ell_0^s$ bounds the critical Hausdorff measure above. □

3 An integer refinement profile

Throughout the proof of the upper bound, $r \geq 2$ is an arbitrary fixed integer. Let

$$P_n(x) = \sum_{i=0}^{n-1} r^{-i} \psi(r^i x).$$

Uniform convergence and positivity give

$$0 \leq T_r(x) - P_n(x) \leq \frac{r^{1-n}}{2(r-1)}. \tag{10}$$

All breakpoints of P_{2m} lie on the grid of equation (2).

Lemma 3.1 (Integral heights and even slopes)**.** *The function P_{2m} is affine on each $I_{m,j}$, its slope there is an even integer, and its endpoint values belong to $\delta_m\mathbb{Z}$.*

Proof. Put $n = 2m$ and $D = 2r^{n-1}$. At an endpoint j/D,

$$DP_n(j/D) = \sum_{d=0}^{n-1} \min\{j \bmod (2r^d),\, 2r^d - (j \bmod (2r^d))\}. \tag{11}$$

Every summand is an integer with the parity of j, and n is even. Thus this sum is even, which proves the height assertion because $D = 2/\delta_m$. On each cell the slope is a sum of $2m$ signs ± 1. □

Write $q = \lfloor r/2 \rfloor$ and $v = \lceil r/2 \rceil$. For an integer z, let $\alpha \in \{0, \ldots, 2r-1\}$ be its residue modulo $2r$, and define

$$W_r(z) = \frac{\min\{\alpha, 2r-\alpha\} + (\alpha \bmod 2)}{2}. \tag{12}$$

It is a $2r$-periodic integer staircase, with

$$W_r(\alpha) = \begin{cases} \lceil \alpha/2 \rceil, & 0 \le \alpha \le r, \\ r - \lfloor \alpha/2 \rfloor, & r \le \alpha \le 2r. \end{cases} \tag{13}$$

In particular $0 \le W_r \le v$ and successive differences belong to $\{-1, 0, 1\}$.

Lemma 3.2 (Two-scale profile)**.** *On a parent $I_{m,j}$, suppose the left endpoint value of P_{2m} is $a\delta_m$ and its slope is $2s$, with $a, s \in \mathbb{Z}$. After subdivision into r^2 children, the dth new endpoint has P_{2m+2}-height, in units of δ_{m+1},*

$$V_d = r^2 a + sd + W_r(r^2 j + d), \qquad 0 \le d \le r^2. \tag{14}$$

On a nonflat parent, P_{2m+2} is monotone. The new two-term contribution lies everywhere between 0 and $v\delta_{m+1}$. If r is even, P_{2m+2} and P_{2m} agree at both parent endpoints.

Proof. Put $n = 2m$ and $z = r^2 j + d$. The new endpoint is $x = z/(2r^{n+1})$. Its old affine height, divided by δ_{m+1}, is $r^2 a + sd$. The two new terms contribute

$$r\psi\left(\frac{z}{2r}\right) + \psi(z/2) = W_r(z),$$

which proves equation (14). Each child is an affine piece, and its endpoint increment is $s + W_r(z + 1) - W_r(z)$. If $s \neq 0$, all such increments have the sign of s or are zero. The bound on the new contribution holds at all fine endpoints by equation (13), and between endpoints by affinity. Finally, for even r, both $r^2 j$ and $r^2(j+1)$ are multiples of $2r$, so the endpoint contributions vanish. □

The two endpoint contributions in an odd base alternate between 0 and $v\delta_{m+1}$. This is the precise place where an adjacent old height strip can contribute to a new crossing.

4 Flat pieces, crossings, and residue separation

For $h \in \mathbb{Z}$, define

$$A_m(h) = \#\{j : P_{2m}|_{I_{m,j}} \equiv h\delta_m\},$$
$$B_m(h) = \#\{j : P_{2m}|_{I_{m,j}} \text{ is nonconstant and } P_{2m}(I_{m,j}) \cap (h\delta_m, (h+1)\delta_m) \neq \varnothing\}.$$

Let $a_m = \max_h A_m(h)$ and $b_m = \max_h B_m(h)$. By lemma 3.1, a nonflat cell counted by $B_m(h)$ contains the whole open strip in its image. All counts refer to cells, including cells in a longer flat interval, rather than to maximal affine pieces.

Lemma 4.1 (Children of a flat parent)**.** *A flat parent at height $a\delta_m$ produces exactly r flat children at each height $(r^2 a + k)\delta_{m+1}$, $1 \le k \le q$. It also produces exactly r nonflat children crossing each fine strip with index $r^2 a + \ell$, $0 \le \ell < v$. There are no other contributions.*

Proof. With $s = 0$, equation (14) follows r consecutive half-periods of W_r. The starting phase is 0 or r modulo $2r$. Each half-period has length r and, by equation (13), has one horizontal step at each height $1, \dots, q$ and one nonhorizontal step across each band $(\ell, \ell+1)$, $0 \le \ell < v$. For even r, the two horizontal steps at the peak lie on opposite sides of a half-period boundary; they are still two distinct cells. The stated counts follow. □

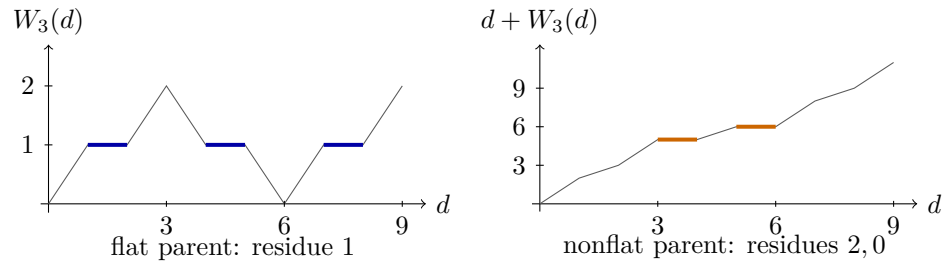


Figure 1: Two ternary local profiles in the even phase, after subtracting the coarse height. Marked horizontal segments occur at height 1 for a flat parent, and at heights $5, 6$ for a parent of half-slope $s = 1$. The incompatible residues persist at every scale.

Lemma 4.2 (Children of a nonflat parent)**.** *A nonflat parent contributes at most one nonflat child to a fixed open fine strip and at most one flat child to a fixed fine height. If it contributes a flat child of height index K, then*

$$K \bmod r \notin \{1, \dots, q\}. \tag{15}$$

Proof. By lemma 3.2, the refined function is monotone on the parent. The integer endpoints of any nonflat child force its image to contain an entire open fine strip as soon as it intersects that strip. Two children in monotone order cannot both contain that same open strip.

A flat child is possible only for $s = 1$ or $s = -1$. Equal-sign nonzero steps of W_r are never adjacent, by equation (13). Hence flat children of this parent are separated by a nonzero increment, so their heights are distinct. To check the residue, put $z = r^2 j + d$ and $\alpha = z \bmod 2r$; then $d \equiv \alpha \pmod r$. For $s = 1$, flatness requires a downward step of W_r, so α is odd and $r \le \alpha \le 2r - 1$. Thus

$$K \equiv d + W_r(z) \equiv (\alpha + 1)/2 \pmod r,$$

whose representative lies in $\{q + 1, \dots, r\}$. For $s = -1$, flatness requires an upward step: α is even and $0 \le \alpha < r$. In that case

$$K \equiv -d + W_r(z) \equiv -\alpha/2 \pmod r.$$

Since $0 \le \alpha/2 \le v - 1$, these residues lie in $\{0, q + 1, \dots, r - 1\}$. This proves equation (15). □

Proposition 4.3 (Coupled recursion)**.** *For $m \ge 1$,*

$$a_{m+1} \le \max\{r a_m, b_m\}, \qquad b_{m+1} \le r a_m + \gamma_r b_m. \tag{16}$$

More precisely, if $K = r^2 h + \ell$ with $0 \le \ell < r^2$, then

$$B_{m+1}(K) \le r A_m(h) \mathbf{1}_{\{\ell < v\}} + B_m(h) + \mathbf{1}_{\{r \text{ odd}\}} B_m(h - 1). \tag{17}$$

Proof. Consider first flat children at height $K\delta_{m+1}$. If $K \bmod r \in \{1, \dots, q\}$, only flat parents contribute, by lemma 4.2. The contribution is $r A_m(h)$ when $1 \le \ell \le q$, and zero otherwise. For the complementary residues, only nonflat parents contribute, each at most once. Here $\ell = 0$ or $\ell \ge q + 1$. On a flat child the old value is the new constant minus a contribution in $[0, v\delta_{m+1}]$. If $\ell = 0$, these old values lie in $((h - 1)\delta_m, h\delta_m]$; since the old function is nonconstant on the child, the parent crosses strip $h - 1$. If $\ell \ge q + 1$, then $v \le q + 1$ gives old values in $[h\delta_m, (h + 1)\delta_m)$, some strictly above $h\delta_m$; the parent crosses strip h. Thus these contributions are bounded by one

of $B_m(h-1), B_m(h)$. The two sources of flat children occupy disjoint residues, proving the first inequality in equation (16).

Now consider a fine open strip of index K. Flat parents contribute $rA_m(h)\mathbf{1}_{\{\ell<v\}}$ by lemma 4.1. Every nonflat parent contributes at most one nonflat child. For odd r, a point in the fine strip has old height in

$$((K-v)\delta_{m+1}, (K+1)\delta_{m+1}) \subset ((h-1)\delta_m, (h+1)\delta_m).$$

The parent's nonconstant affine image has integer grid endpoints, so it crosses strip $h-1$ or strip h. Their union has at most $B_m(h-1)+B_m(h)$ parents. For even r, the refined function is monotone on a nonflat parent and keeps its two endpoint values. Its image therefore equals the old affine image. The fine strip lies in the old open strip h, so only $B_m(h)$ parents are needed. This proves equation (17); taking maxima gives the second inequality in equation (16). □

Proposition 4.4 (Explicit counting bounds)**.** *For every $m \geq 1$,*

$$a_m = 2r^{m-1}, \qquad b_m \leq \frac{2(r^m - \gamma_r^m)}{r-\gamma_r}. \tag{18}$$

Proof. The two half-periods of P_2 show $a_1 = b_1 = 2$. Because $r - \gamma_r \geq 1$, the bounds at step m imply

$$a_{m+1} \leq \max\left\{2r^m, \frac{2(r^m-\gamma_r^m)}{r-\gamma_r}\right\} \leq 2r^m,$$
$$b_{m+1} \leq 2r^m + \frac{2\gamma_r(r^m-\gamma_r^m)}{r-\gamma_r} = \frac{2(r^{m+1}-\gamma_r^{m+1})}{r-\gamma_r}.$$

This proves the upper bounds by induction. Starting with the two flat cells at height $1/r$ and always choosing label $k=1$ in lemma 4.1 gives $2r^{m-1}$ flat cells at a common height. Thus the bound for a_m is exact. □

The invariant behind this induction is also useful conceptually. If $c = r/(r-\gamma_r)$, then $c \leq r$ and the bound $(a,b) \leq P(1,c)$ is sent by equation (16) to at most $rP(1,c)$. In particular, the extra odd-base coefficient 2 remains compatible with the covering rate r, even at the smallest odd base 3.

5 Actual level sets and maximal dimension

Lemma 5.1 (A tail-sensitive cover)**.** *For every $y \in \mathbb{R}$, with $h = \lfloor y/\delta_m \rfloor$,*

$$N_{r,2m}(y) \leq A_m(h) + B_m(h-1) + B_m(h). \tag{19}$$

Proof. By equation (10), a point with $T_r(x) = y$ has

$$P_{2m}(x) \in [y - \eta\delta_m, y], \qquad \eta = \frac{1}{2(r-1)} < 1.$$

This interval contains at most one grid height, necessarily $h\delta_m$. Thus flat cells are bounded by $A_m(h)$. For a nonflat cell, write its image endpoints as $a\delta_m, b\delta_m$ with integers $a < b$. If its image meets that interval, then $a \leq y/\delta_m < h+1$ and $b \geq y/\delta_m - \eta > h-1$. Hence $a \leq h$ and $b \geq h$. If $b = h$, it crosses strip $h-1$; if $b \geq h+1$, it crosses strip h. This includes contacts at closed cell endpoints and ordinates that are themselves grid heights. □

Proof of theorem 1.1. Combine equations (18) and (19) to obtain the first bound in equation (3). Since $r/(r-\gamma_r) \le 3$, the second bound follows. The participating cells cover $L_r(y)$ and have length $\ell_m = 1/(2r^{2m-1})$. Their number is $O(r^m)$, so

$$\limsup_{m\to\infty} \frac{\log \mathcal{N}_{\ell_m}(L_r(y))}{-\log \ell_m} \le \frac{1}{2}.$$

Successive lengths have fixed ratio r^2; using the next finer cover between these lengths proves the full upper box bound. The Hausdorff bound follows. Finally, the critical Hausdorff covering cost is at most $N_{r,2m}(y)\sqrt{\ell_m}$. Taking the limit in the explicit first bound of equation (3) gives the measure estimate in equation (4). □

5.1 Flat-return families

Call a half-r-adic cell I *flat of level* n if P_n is constant on I, where $n \ge 1$ and I has length $1/(2r^{n-1})$. Its level n is necessarily even. If $P_n|_I = c$, then lemma 4.1 gives, for each $k \in \{1,\dots,q\}$, exactly r flat two-step descendants at height $c + k/r^{n+1}$.

Proposition 5.2 (Maximal-dimensional height codes)**.** *For every such I, c and every sequence $k_i \in \{1,\dots,q\}$, put*

$$y = c + \sum_{i=1}^{\infty} \frac{k_i}{r^{n+2i-1}}. \tag{20}$$

Then $I \cap L_r(y)$ contains a compact set K with $0 < \mathcal{H}^{1/2}(K) < \infty$. Consequently $\dim_{\mathrm{H}} L_r(y) = 1/2$. For fixed I, the set of coded ordinates has dimension $\log q/(2\log r)$, interpreted as zero when $q = 1$.

Proof. At step i, retain all r flat two-step children of label k_i in every retained interval. After m steps there are r^m intervals of length $|I|r^{-2m}$, with disjoint interiors, on which P_{n+2m} has the common value $c + \sum_{i=1}^{m} k_i/r^{n+2i-1}$. The tail tends uniformly to zero, so their nested intersection is contained in $L_r(y)$. Apply lemma 2.1 with $b = r$ and $\theta = r^{-2}$, and then theorem 1.1. The ordinates in equation (20) are an affine image of $\{\sum_{i\ge1} k_i(r^2)^{-i} : 1 \le k_i \le q\}$. For $q \ge 2$, its q first-level interval hulls have disjoint interiors and a positive gap, so the same lemma gives the stated dimension. For $q = 1$ it is a singleton. □

Proof of the first assertions of theorem 1.2. Take $I = [(2k-1)/(2r), k/r]$, where $P_2 = k/r$, and keep the constant label k. Formula equation (20) sums to $kr/(r^2-1)$, giving equation (5).

Every nonempty open subinterval U of $[0,1]$ contains a flat cell. Indeed choose a sufficiently small half-r-adic cell inside U. Its partial-sum slope is an integer. The next summand has both derivative signs among its r children; select the sign that reduces the absolute slope by one. Repeating reaches zero after finitely many steps. Apply proposition 5.2 inside that flat cell. Shrinking U to an arbitrary point and using continuity of T_r proves density of the coded maximal-dimensional ordinates throughout its range. The irrational assertions are proved in section 7. □

6 Odd bases: exact ranges and arithmetic rigidity

In this section $r = 2q+1 \ge 3$ is odd. Put $A = r/(r-1)$ and define the half-graph and its integer tilts by

$$f(u) = 2T_r(u/2), \qquad f_t(u) = f(u) + tu \quad (0 \le u \le 1, t \in \mathbb{Z}).$$

Since $r^n/2$ is a half-integer, $T_r(1/2) = A/2$. The termwise upper bound in equation (10) shows $M_r = A/2$. Also $T_r(1-x) = T_r(x)$. The identity $\psi(v+1/2) = 1/2 - \psi(v)$ gives

$$f(u) + f(1-u) = A. \tag{21}$$

Let $\tau : [0,1] \to [0,1]$ be the continuous tent map with r laps, whose inverse branches are

$$h_j(v) = \frac{b_j + \varepsilon_j v}{r}, \quad b_j = j + (j \bmod 2), \quad \varepsilon_j = (-1)^j, \qquad 0 \le j < r. \tag{22}$$

Folding $ru/2$ by period one and reflection in the first-term identity for T_r gives $f(u) = u + r^{-1} f(\tau u)$. Therefore

$$f_t(h_j(v)) = z \quad \Longleftrightarrow \quad f_{\varepsilon_j(t+1)}(v) = rz - (t+1)b_j. \tag{23}$$

We call (t, z) a *state*, and call it admissible if z lies in the range of f_t. Each inverse word gives a closed r-adic interval; its orientation can be reversed.

Lemma 6.1 (Exact tilted ranges)**.** *For integer t, the range $R_t = f_t([0,1])$ is*

$$R_t = \begin{cases} [0, A+t], & t \ge 0, \\ [A - s - A/r^s,\ A/r^s], & t = -s,\ s \ge 1. \end{cases} \tag{24}$$

Proof. For $t \ge 0$, the bounds $0 \le f \le A$ and the endpoint values give the answer by continuity. For $s \ge 1$, the first s summands of $f(u)$ are each at most u, and the remaining tail is at most A/r^s. Thus $f(u) - su \le A/r^s$, with equality at $u = r^{-s}$. Indeed the first s summands equal u there and the remaining half-graph equals $r^{-s} f(1)$. Finally, equation (21) implies $f_{-s}(u) + f_{-s}(1-u) = A - s$, which determines the minimum. Continuity supplies every intermediate value. □

Fix $1 \le k \le q$ for the rest of this section, and put

$$D = r^2 - 1, \quad d = \frac{2k}{D}, \quad z_* = rd, \quad S = (d + 2\mathbb{Z}) \cup (z_* + 2\mathbb{Z}). \tag{25}$$

Since $r^2 d = d + 2k$ and b_j is even, all child states of a state with $z \in S$ again have height in S.

Lemma 6.2 (Separation of residues and overlaps)**.** *Among admissible states with $z \in S$, precisely the following two have more than one admissible child:*

$$(0, z_*) \longrightarrow \begin{cases} (1, 2+d) & j = 2k-2, \\ (-1, d) & j = 2k-1, \\ (1, d) & j = 2k, \end{cases} \tag{26}$$

$$(-1, d) \longrightarrow (0, z_*) \quad \textit{for every } 0 \le j < r. \tag{27}$$

Every other admissible state with height in S has exactly one admissible child and cannot reach either of these two states. Its solution set is a singleton. No solution in any of these states is an endpoint of an r-adic interval at any finite depth.

Proof. We first exclude multiple children when $a = t+1$ has $|a| \ge 2$. The endpoint heights of rf_t on the jth branch, in spatial order, are

$$\begin{cases} (aj,\ a(j+1) + A), & j \text{ even}, \\ (aj + A,\ a(j+1)), & j \text{ odd}. \end{cases}$$

They form a strictly monotone sequence, in the sign of a, with successive gaps at least $|a| - A$. By equation (24), the image of a branch stays within its endpoint interval if its child tilt is positive; if that tilt is negative, it extends beyond each endpoint by exactly $B = A/r^{|a|}$. Since

$$2B \le \frac{2}{r(r-1)} \le \frac{1}{3} < \frac{1}{2} \le |a| - A,$$

nonadjacent branch images are disjoint. An overlap of adjacent images lies within distance B of their common endpoint height. Such a height belongs to $2\mathbb{Z} \cup (A + \mathbb{Z})$, because even-indexed spatial endpoints have height aj with even j, and odd-indexed ones have height $aj + A$.

On the other hand,

$$\operatorname{dist}(\{d, z_*\},\, 2\mathbb{Z} \cup (A + \mathbb{Z})) \ge \frac{2}{D} > \frac{A}{r^2}. \tag{28}$$

Here $0 < d \le 1/(r+1) < A - 1 < z_* \le r/(r+1) < 1$, and the potentially closest distances satisfy

$$d \ge \frac{2}{D}, \quad (A-1) - d \ge \frac{2}{D}, \quad z_* - (A-1) \ge \frac{r-1}{D}, \quad A - z_* \ge \frac{2r}{D}.$$

All other lattice points are farther away. The same bound holds after adding an even integer. Since $rz \in S$ for $z \in S$, it excludes every overlap described above. Admissibility ensures at least one child, so there is exactly one when $|t+1| \ge 2$.

The remaining possibilities are $t = 0, -1, -2$. The ranges in equation (24) show

$$R_0 \cap S = \{d, z_*\}, \qquad R_{-1} \cap S = \{d\}, \qquad R_{-2} \cap S = \varnothing.$$

For the last assertion, $R_{-2} = [-(r-1)/r, 1/(r(r-1))]$, whereas positive elements of S exceed its upper endpoint and negative elements are less than -1. Substitution in equation (23) gives equations (26) and (27); the remaining state $(0, d)$ has the single child $(1, z_*)$ at $j = 0$.

A child of tilt 0 requires a parent of tilt -1, whose only admissible height in S is d. A child of tilt -1 requires a parent of tilt 0 or -2; the latter is impossible, and $(0, d)$ does not produce tilt -1. Thus no state outside the displayed pair can enter it. Such a state has exactly one surviving interval at every depth. These intervals are nested, their lengths tend to zero, and the solution set is nonempty by admissibility; it is therefore a singleton. Finally an endpoint solution at any depth would require a local height 0 or $A + t$. Both are excluded by $z \in S$ and equation (28). This also prevents double counting at every branch boundary. □

6.1 The entire critical level set

Let $E_* = \{u \in [0,1] : f(u) = z_*\}$. The middle branch in equation (26) returns to E_* after two steps, while the other two branches each give one point. Let S_0 denote these two points and put

$$F_j = h_{2k-1} \circ h_j \quad (0 \le j < r), \qquad E_* = S_0 \;\cup \bigcup_{j=0}^{r-1} F_j(E_*). \tag{29}$$

The two points of S_0 lie strictly inside branches $2k-2$ and $2k$: the corresponding targets $2 + d$ and d lie strictly between the endpoint values $0, A + 1$ of f_1. Thus S_0 is disjoint from $J = h_{2k-1}([0,1]) = [(2k-1)/r, 2k/r]$. The maps F_j have ratio r^{-2}, and their interval images partition J, up to shared endpoints.

Proposition 6.3 (Cantor set and isolated points)**.** *Let K be the attractor of the maps F_j, equivalently the intersection of their nested word-interval unions. Then*

$$E_* = K \ \cup \bigcup_w F_w(S_0), \qquad 0 < \mathcal{H}^{1/2}(K) < \infty, \tag{30}$$

where $w = (w_1, \dots, w_n)$ ranges over all finite words on $\{0, \dots, r-1\}$, $F_w = F_{w_1} \circ \cdots \circ F_{w_n}$, and the empty word denotes the identity map. The union outside K consists of isolated points and the set of accumulation points of E_* is exactly K. Consequently

$$L_r(kr/D) = \tfrac{1}{2}E_* \ \cup \ (1 - \tfrac{1}{2}E_*)$$

has the structure asserted in theorem 1.3.

Proof. Iterating equation (29), a point either reaches S_0 at a finite step or remains in nested word intervals and belongs to K. Conversely, every $F_w(S_0)$ is contained in E_* by the same equation. For a point of K, choose any $p \in S_0$ and a nested address w_n for the point. Then $F_{w_n}(p) \in E_*$ converges to it, so it belongs to the closed set E_*. This proves the representation.

The uniform interval construction has r children and contraction r^{-2}, so lemma 2.1 gives the measure assertion. The compact interval J lies in $(0, 1)$; each $F_j(J)$ therefore lies strictly inside its respective word interval. This gives positive gaps between the retained pieces. Hence K is a Cantor set: it is compact, perfect (each address has arbitrarily deep alternative children), and nowhere dense (its step-n cover has total length r^{-n}). The two points in S_0 are isolated because all other points of E_* lie in J. Also $E_* \subset (0, 1)$, so the compact sets $F_j(E_*)$ lie strictly inside their distinct branch intervals. Repeated use of equation (29) proves that every $F_w(S_0)$ is isolated and is outside K. The approximating points $F_{w_n}(p)$ above are outside K; thus every point of K is an accumulation point. All others are isolated. The two copies in the displayed formula are disjoint, since neither endpoint 1 nor 0 belongs to E_*. □

For completeness, the same states give exact covering counts, without counting merely the Cantor subset. Let C_n be the number of depth-n r-adic cells meeting E_*. By lemma 6.2 there are no boundary solutions. The two singleton branches each contribute one cell at every subsequent depth, and the other branch returns through r copies in two steps. Therefore

$$C_0 = 1, \quad C_1 = 3, \quad C_{n+2} = rC_n + 2,$$

and induction gives

$$C_{2m} = \frac{(r+1)r^m - 2}{r-1}, \qquad C_{2m+1} = \frac{(3r-1)r^m - 2}{r-1}. \tag{31}$$

There are two reflected halves of $L_r(kr/D)$, each subdivided to depth $2m-1$ in equation (2). Thus $N_{r,2m}(kr/D) = 2C_{2m-1}$, proving equation (8).

6.2 A finite classification on the arithmetic family

Suppose $y \in \mathcal{A}_{r,k}$ and m witnesses equation (7). Start with $(0, 2y)$ and expand all admissible branches for m steps using equations (23) and (24). Every final height has the form $r^m(2y) - B$ with $B \in 2\mathbb{Z}$, and hence belongs to S. If any surviving state is $(0, z_*)$ or $(-1, d)$, its local solution set contains a similarity copy of K from proposition 6.3; thus $\mathcal{H}^{1/2}(L_r(y)) > 0$. The upper bound is theorem 1.1. If neither state occurs, every final state has a singleton solution set by lemma 6.2, so

the entire level set is finite. This is a finite procedure with at most $1 + r + \cdots + r^m$ nodes and only rational arithmetic. Its output is independent of the chosen witness, because it describes the actual solution sets in a finite interval cover. Finally the numbers $(h + k/D)/r^m$, with $h \in \mathbb{Z}$, form meshes of size r^{-m} contained in the defining residue class. Restricting to $[0, M_r]$ and letting $m \to \infty$ proves density of $\mathcal{A}_{r,k}$. Together with proposition 6.3 and equation (31), this proves theorem 1.3.

Corollary 6.4 (An explicit ternary condensation set)**.** *Define* $G_d(x) = x/9 + d/18$ *for* $d \in \{3, 4, 5\}$*, and set*

$$K_3 = \left\{ \frac{3}{16} + \sum_{i=1}^{\infty} \frac{e_i}{2 \cdot 9^i} : e_i \in \{0, 1, 2\} \right\}.$$

There is a compact set X *satisfying*

$$X = \{17/108, 37/108\} \ \cup \ \bigcup_{d=3}^{5} G_d(X), \tag{32}$$

$$L_3(3/8) = X \cup (1 - X). \tag{33}$$

Its explicit representation is $X = K_3 \cup \bigcup_w G_w(\{17/108, 37/108\})$*. The accumulation set of* $L_3(3/8)$ *is* $K_3 \cup (1 - K_3)$*; every other point is isolated.*

Proof. Here $d = 1/4$ and $z_* = 3/4$. The state $(1, 1/4)$ follows the unique orbit

$$(1, 1/4) \xrightarrow{0} (2, 3/4) \xrightarrow{0} (3, 9/4) \xrightarrow{1} (-4, -5/4) \xrightarrow{1} (3, 9/4).$$

The repeated branch $h_1(v) = (2 - v)/3$ has fixed point $1/2$. Hence the unique solution in the first state is $(1/2)/3^2 = 1/18$. By equation (21), $f_1(u) + f_1(1 - u) = 5/2$, so the solution of $f_1(u) = 9/4$ is $17/18$. The two outside points of E_* are therefore $17/54$ and $37/54$. The set E_* is invariant under $u \mapsto 1 - u$, again by equation (21). Consequently the three reflected maps $h_1 \circ h_j$ acting on E_* may be replaced by the positive maps $u \mapsto (u + d)/9$, $d = 3, 4, 5$. Taking $X = E_*/2$ proves equation (32); the positive-map attractor is exactly K_3 by summing its base-nine address expansion. The last assertions follow from proposition 6.3. □

The isolated points $17/108$ and $37/108$ already appear in [3, Remark 4.7]. The description above accounts for their entire similarity orbits and the remaining perfect part; it does not claim those two points as new observations.

7 Sparse four-step changes and irrational ordinates

Two-step flat codes give only one ordinate per initial flat interval when $r = 3$, since $q = 1$. To obtain irrational ordinates without losing dimension, we combine four-step returns with two different branching numbers and use the slower return only at a set of times of density zero.

Lemma 7.1 (Two ternary four-step returns)**.** *Let* I *be a flat level-*n *cell for* P_n*, where* n *is even, and let* $P_n|_I = c$*. Subdivide* I *into* 81 *equal cells, indexed from* 0 *to* 80*. There are three flat* P_{n+4} *cells at height* $c + 5/3^{n+3}$*, and nine at height* $c + 10/3^{n+3}$*. The indices are given in table 1; the phase is the residue of* $3^n \inf I$ *modulo one, either* 0 *or* $1/2$*.*

Phase	Increment label 5	Increment label 10
0	$3, 50, 57$	$10, 13, 16, 37, 40, 43, 64, 67, 70$
$1/2$	$23, 30, 77$	$10, 13, 16, 37, 40, 43, 64, 67, 70$

Table 1: Four-step flat returns, indexed within a parent. All selected cells of either label have disjoint interiors and positive gaps.

Proof. The new contribution is $3^{-n}P_4(3^n x)$ and the new variable traverses an interval of length $3/2$. Let $H_j = 27P_4(j/54)$. Formula equation (11) gives the exact integer expression

$$H_j = \frac{1}{2}\sum_{a=0}^{3} \min\{j \bmod (2\cdot 3^a),\ 2\cdot 3^a - (j \bmod (2\cdot 3^a))\}.$$

For $0 \le j \le 27$ its values, in order, are

$$(0, 2, 3, 5, 5, 6, 6, 8, 9, 11, 10, 10, 9, 10,\\ 10, 11, 10, 10, 9, 11, 12, 14, 14, 15, 15, 17, 18, 20).$$

The adjacent equal pairs have left indices $3, 5, 10, 13, 16, 21, 23$, with heights $5, 6, 10, 10, 10, 14, 15$. Reflection $P_4(1-u) = P_4(u)$ and period one determine all indices in the phase-zero interval $[0, 3/2]$. In the other phase use $27P_4(u+1/2) = 20 - 27P_4(u)$; this follows termwise from $\psi(v+1/2) = 1/2 - \psi(v)$ and $27\max P_4 = 20$. These formulas give exactly table 1. They also show that the full list of flat labels in either phase has multiplicities $5:3$, $6:3$, $10:9$, $14:3$, $15:3$. □

Proof of the remaining assertions of theorem 1.2. Start with $I = [1/6, 1/3]$, on which $P_2 = 1/3$. For $E \subset \mathbb{N}$, at four-step stage j choose label 5 if $j \in E$ and label 10 otherwise, retaining all cells of the chosen label in every parent. The common limiting height is

$$\frac{1}{3} + \sum_{j\ge 1} \frac{D_j}{3^{4j+1}} = \frac{3}{8} - \frac{5}{3}\sum_{j\in E} 81^{-j} = y_E, \qquad D_j = \begin{cases} 5, & j \in E, \\ 10, & j \notin E. \end{cases}$$

Uniform convergence shows that the nested intersection lies in $L_3(y_E)$. If $e_m = \#(E \cap \{1, \dots, m\})$, its step-$m$ intervals number 3^{2m-e_m} and have length $|I|81^{-m}$. By lemma 2.1, their intersection has dimension

$$\liminf_{m\to\infty} \frac{(2m - e_m)\log 3}{4m\log 3 - \log|I|}.$$

This is $1/2$ if E has asymptotic density zero, and theorem 1.1 gives equality for the entire level set.

Moreover $3y_E - 1 = \sum_{j\ge 1} D_j 81^{-j}$ is a base-81 expansion using digits 5 and 10; it has no terminating-expansion ambiguity. A rational number has an eventually periodic base-81 expansion: long division has only finitely many possible remainders. Conversely every eventually periodic expansion sums to a rational number. If E is an infinite subset of the squares, its indicator cannot be eventually periodic, since such a set would have either positive density or only finitely many elements. Hence y_E is irrational. For two different sets, the contribution at the first differing index has magnitude $(5/3)81^{-j}$, while the total possible later contribution is at most $1/80$ of that magnitude. Their heights are distinct. There are uncountably many infinite subsets of the squares (the finite subsets form a countable subfamily of its power set), proving the final assertion. □

In particular the explicit irrational height

$$y_{\mathrm{sq}} = \frac{3}{8} - \frac{5}{3}\sum_{j=1}^{\infty} 81^{-j^2}$$

has $\dim_{\mathrm{H}} L_3(y_{\mathrm{sq}}) = 1/2$. The construction explains why irrationality is compatible with the sharp bound: changing the height infinitely often is inexpensive when the number of changes is negligible compared with the number of scales.

8 Integer linear perturbations

Corollary 8.1. *For every integer $r \geq 2$ and every integer t,*

$$\max_{z\in\mathbb{R}} \dim_{\mathrm{H}}\{x \in [0,1] : T_r(x) + tx = z\} = \frac{1}{2}.$$

Every such level set also has upper box dimension at most $1/2$ and finite $1/2$-dimensional Hausdorff measure.

Proof. For $k \geq 0$ and $0 \leq u \leq 1$, the first k triangular terms are linear at u/r^k, giving

$$T_r(u/r^k) = r^{-k}\big(T_r(u) + ku\big). \tag{34}$$

Thus a fiber of $T_r(u) + ku$ is similar to a subset of an ordinary T_r level set. Theorem 1.1 proves all the upper assertions. Conversely,

$$T_r(1 - u/r^k) + k(1 - u/r^k) = k + r^{-k}T_r(u).$$

The image of any level set from equation (5) therefore gives a dimension-$1/2$ subset of a fiber with slope k. Finally reflection changes slope $-k$ to k, up to an additive constant: $T_r(1-u) - k(1-u) = T_r(u) + ku - k$. □

9 Exact cross-checks and the supplementary script

The accompanying Python 3 file is named `takagi_level_set_checks.py`. It uses only the standard library; all decisions use integers or `fractions.Fraction`. Running

```
python3 takagi_level_set_checks.py
```

runs table 2 and stops on a failed assertion. All comparisons are exact.

For the polygon checks the maximal m is $6, 5, 4, 4, 3, 3, 3, 3$ for $r = 2, 3, 4, 5, 6, 7, 8, 9$, respectively. The residue checks use $r \in \{3, 5, 7, 9, 11, 13, 15\}$, all $1 \leq k \leq (r-1)/2$, $-20 \leq t \leq 20$, and the admissible heights $d + 2h$, $z_* + 2h$ with $-12 \leq h \leq 12$. The critical-count check uses all such k for $r \in \{3, 5, 7, 9, 11\}$. These ranges describe cross-checks, not hypotheses of any theorem. All universal statements are proved by the preceding arguments. The script is neither a formal proof in a proof assistant nor an independent verification of its own implementation.

Script function	Exact object checked	Text connection
`check_local_profiles`	All $2 \le r \le 41$, half-slopes $-6 \le s \le 6$, both possible phases	lemmas 4.1 to 4.2
`check_global_polygons`	All polygonal cells for the finite base/depth pairs specified below	propositions 4.3 and 4.4
`check_ternary_tail_cover`	Every real ordinate at $r = 3$, $1 \le m \le 4$, using exact infinite-function image ranges	lemma 5.1
`check_residue_rigidity`	Finite odd-base tilted states with heights in S	lemma 6.2
`check_critical_counts`	Critical counts through depth 24; the ternary singleton orbit	equation (31) and corollary 6.4
`check_sparse_equality_blocks`	All 81 four-step cells in both phases	lemma 7.1
`check_scalar_counterexample`	The exact finite counts in equation (35)	One-count obstruction below

Table 2: The role and finite scope of each supplementary check.

9.1 Two exact enumeration procedures

The polygon procedure below computes all A_m and B_m directly from equation (11), without using their proposed recursions. Its integer event sweep includes every open strip crossed by a nonflat cell.

Listing 1: Direct polygon counts (function polygon_counts).

```
PolygonCounts(r, m):
    n = 2*m; J = 2*r^(n-1)
    for j = 0,...,J:
        H[j] = (1/2)*sum_{d=0}^{n-1}
                  min(j mod (2*r^d), 2*r^d - (j mod (2*r^d)))
    A = zero counter; events = zero counter
    for j = 0,...,J-1:
        if H[j] == H[j+1]: A[H[j]] += 1
        else:
            lo = min(H[j], H[j+1]); hi = max(H[j], H[j+1])
            events[lo] += 1; events[hi] -= 1
    active = 0; B = zero counter
    for h = min(H),...,max(H):
        active += events[h]; B[h] = active
    return A, B
```

The second procedure counts cells for the actual infinite half-graph. The exact range test is equation (24), not a truncated-tail estimate. It applies here to odd r. Multiplicities of coincident states must be retained because different cells can lead to the same state.

Depth 7			Depth 9		
t	z	Multiplicity	t	z	Multiplicity
-1	$1/4$	5	-1	$1/4$	15
1	$1/4$	5	1	$1/4$	15
5	$25/4$	3	1	$9/4$	15
7	$33/4$	2	3	$9/4$	5
			7	$25/4$	3
			9	$33/4$	2

Table 3: Exact states witnessing failure of the scalar recursion.

Listing 2: Exact state expansion (functions children and advance).

```
Expand(r, t, z, depth):
    if z is not in R_t: return empty counter
    states = counter with multiplicity 1 at (t,z)
    repeat depth times:
        next = zero counter
        for (t,z), multiplicity in states:
            for j = 0,...,r-1:
                s = (-1)^j * (t+1)
                w = r*z - (t+1)*(j + (j mod 2))
                if w is in R_s:
                    next[(s,w)] += multiplicity
        states = next
    return states
```

Each inverse word is a distinct closed cell, so the sum of the returned multiplicities is its covering count, with the stated boundary convention. For the ternary tail-cover check, the code separately generates the exact image interval of every word cell. It then sweeps the sorted image endpoints, processing starts before ends at a common height, and also inserts every coarse grid height. Between successive event heights both the actual count and the proposed bound are constant. Testing the event heights and one interior point of each gap therefore checks every real ordinate at those finite depths, including endpoint-only coincidences.

9.2 Why a single total-count recursion is insufficient

Counting only the pieces already known to be flat omits branches that return to flatness later. Even a count of all surviving cells does not satisfy the tempting uniform recursion $N_{3,2m+2}(y) \le 3N_{3,2m}(y)+4$. For the exact rational ordinate

$$y = \frac{6425}{17496}, \qquad N_{3,8}(y) = 30, \qquad N_{3,10}(y) = 110, \qquad 110 > 3 \cdot 30 + 4. \tag{35}$$

For reproducibility, table 3 gives the states returned by the second procedure at depths 7 and 9 from $(0, 2y)$. Their total multiplicities are 15 and 55; the two reflected halves give the displayed counts. The supplementary script reproduces both full counters. This finite example is illustrative and is not an input to the upper bound. The recurrence being disproved here is a possible shortcut, not a claim attributed to the cited authors.

The coupled proof succeeds because it retains two distinct types of geometric information and then uses their height residues. Without that separation, adding the possible sources of flat pieces

leads to a larger growth estimate. The useful conclusion is not that finite block enumeration is invalid; it is that the local residue exclusion supplies information a single total count does not record.

10 Further questions

The uniform bound leaves a finer classification problem: which ordinates have dimension 1/2, and among them which have positive critical Hausdorff measure? The arithmetic rigidity in section 6 and the sparse changes in section 7 describe two mechanisms, but do not exhaust all possible return patterns. Extending the classification beyond the dense arithmetic classes would require controlling nonperiodic residue orbits rather than only a two-residue cycle.

The distinction between dimension and critical measure is substantive. If E in equation (6) is infinite, the retained subset used in the proof has zero 1/2-dimensional measure: its step-m cover has 1/2-cost $|I|^{1/2}3^{-e_m} \to 0$. This does not determine the measure of the entire level set, because other branches may contribute. Likewise the integer-slope reduction uses exact scaling and does not settle arbitrary real linear perturbations.

Disclosure of AI assistance

ChatGPT (OpenAI) was used extensively in the research dialogue underlying this manuscript, including literature exploration, proposing and developing mathematical arguments, generating and debugging exact-arithmetic code, and drafting and revising the exposition. In particular, the coupled counting proof and the residue-separation arguments were developed with AI assistance. The supplementary script was generated and executed in that workflow. Its finite checks do not constitute formal verification or independent peer review. The AI system is not a named author. Responsibility for the final manuscript and its mathematical claims rests with the author.

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